

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

“JnanaSangama”, Belgaum -590014, Karnataka.



LAB REPORT on

Database Management Systems (23CS3PCDBM)

Submitted by

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in partial fulfillment for the award of the degree of
BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING



B.M.S. COLLEGE OF ENGINEERING

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B. M. S. College of Engineering,
Bull Temple Road, Bangalore 560019
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Department of Computer Science and Engineering



CERTIFICATE

This is to certify that the Lab work entitled “Database Management Systems (23CS3PCDBM)” carried out by **HITISH RAO P (1BM23CS116)**, who is bonafide student of **B. M. S. College of Engineering**. It is in partial fulfillment for the award of **Bachelor of Engineering in Computer Science and Engineering** of the Visvesvaraya Technological University, Belgaum during the year 2022. The Lab report has been approved as it satisfies the academic requirements in respect of a Database Management Systems (23CS3PCDBM) work prescribed for the said degree.

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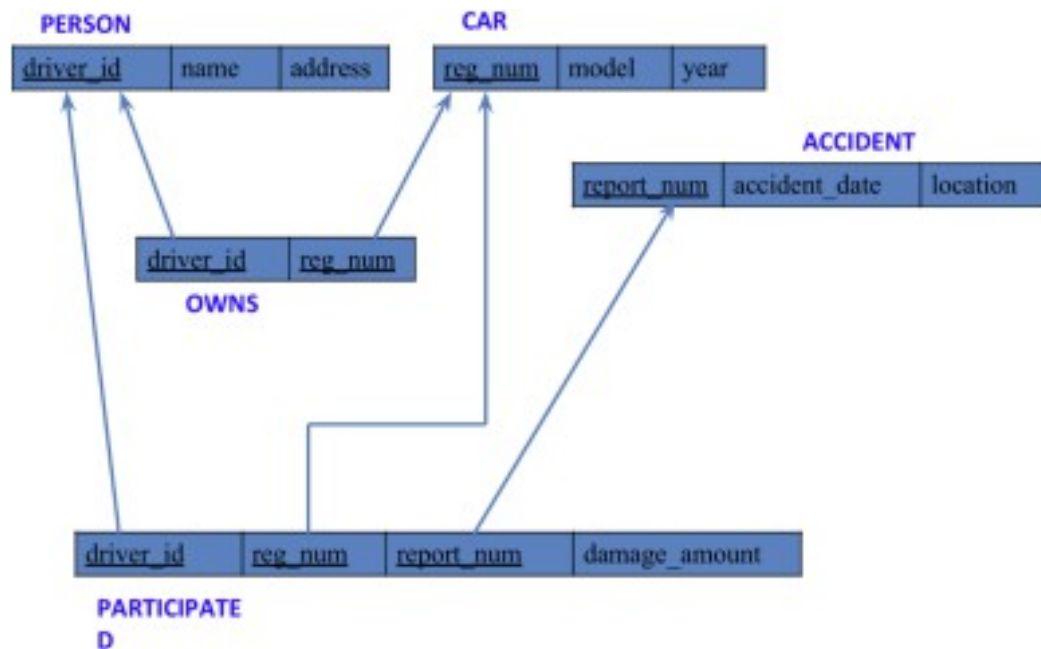
Insurance Database

Question

(Week 1)

- PERSON (driver_id: String, name: String, address: String)
- CAR (reg_num: String, model: String, year: int)
- ACCIDENT (report_num: int, accident_date: date, location: String)
- OWNS (driver_id: String, reg_num: String)
- PARTICIPATED (driver_id: String, reg_num: String, report_num: int, damage_amount: int)
- Create the above tables by properly specifying the primary keys and the foreign keys. - Enter at least five tuples for each relation
- Update the damage amount to 25000 for the car with a specific reg_num (example 'KA053408') for which the accident report number was 12.
- Add a new accident to the database.
- To Do
 - Display Accident date and location
- Display driver id who did accident with damage amount greater than or equal to Rs.25000

Schema Diagram



Create database

```
create database Hitish_116;
use Hitish_116;
```

Create table

```
create table person (
  driver_id varchar(10),
  name varchar(20),
  address varchar(20),
  primary key(driver_id)
);
```

```
create table car (
  reg_num varchar(10),
  model varchar(10),
  year int,
  primary key (reg_num)
);
```

```
create table accident(
  report_num int,
  accident_date date,
  location varchar(20),
  primary key (report_num)
);
```

```

create table owns(
driver_id varchar(10),
reg_num varchar(10),
primary key (driver_id , reg_num),
foreign key(driver_id) references person(driver_id),
foreign key(reg_num) references car(reg_num)
);

```

```

create table participated (
driver_id varchar(10),
reg_num varchar(10),
report_num int,
damage_amount int,
primary key (driver_id, reg_num, report_num),
foreign key (driver_id) references person(driver_id),
foreign key (reg_num) references car(reg_num),
foreign key (report_num) references accident(report_num)
);

```

Structure of the table

desc person;

Result Grid						
		Filter Rows:			Export:	Wrap Cell Content:
	Field	Type	Null	Key	Default	Extra
▶	driver_id	varchar(10)	NO	PRI	NULL	
	name	varchar(20)	YES		NULL	
	address	varchar(20)	YES		NULL	

desc car;

Result Grid						
		Filter Rows:			Export:	Wra
	Field	Type	Null	Key	Default	Extra
▶	reg_num	varchar(10)	NO	PRI	NULL	
	model	varchar(10)	YES		NULL	
	year	int	YES		NULL	

desc accident;

Result Grid						
		Filter Rows:			Export:	Wrap Cell Content:
	Field	Type	Null	Key	Default	Extra
▶	report_num	int	NO	PRI	NULL	
	accident_date	date	YES		NULL	
	location	varchar(20)	YES		NULL	

desc owns;

Result Grid						
		Filter Rows:	Export:		Wrap Cel	
	Field	Type	Null	Key	Default	Extra
▶	driver_id	varchar(10)	NO	PRI	NULL	
	reg_num	varchar(10)	NO	PRI	NULL	

desc participated;

	Field	Type	Null	Key	Default	Extra
▶	driver_id	varchar(10)	NO	PRI	NULL	
	reg_num	varchar(10)	NO	PRI	NULL	
	report_num	int	NO	PRI	NULL	
	damage_amount	int	YES		NULL	

Inserting Values to the table

insert into person

```
values('a01','richard','srinivasnagar'),  
( 'a02','pradeep','rajajinagar'),  
( 'a03','smith',' ashoknagar'),  
( 'a04','venu','NRcolony'),  
( 'a05','jhon','hanumanthnagar');
```

	driver_id	name	address
▶	a01	richard	srinivasnagar
	a02	pradeep	rajajinagar
	a03	smith	ashoknagar
	a04	venu	NRcolony
	a05	jhon	hanumanthnagar

insert into car

```
values ('KA052250', 'indica', 1990),  
( 'KA031181', 'lancer', 1957),  
( 'KA095477', 'toyota', 1998),  
( 'KA053408', 'honda', 2008),  
( 'KA041702', 'audi', 2005);
```

	reg_num	model	year
▶	KA031181	lancer	1957
	KA041702	audi	2005
	KA052250	indica	1990
	KA053408	honda	2008
	KA095477	toyota	1998

insert into accident

```
values(11,'2001-01-03','mysoreroad'),  
(12,'2002-02-04','southendcircle'),
```

```
(13,'2021-01-03','bulltempleroad'),
(14,'2017-02-08','mysoreroad)'),
(15,'2004-03-05','kanakpuraroad');
```

	report_num	accident_date	location
▶	11	2001-01-03	mysoreroad
	12	2002-02-04	southendcircle
	13	2021-01-03	bulltempleroad
	14	2017-02-08	mysoreroad)
	15	2004-03-05	kanakpuraroad

insert into owns

```
values('a01','KA052250'),
('a02','KA053408'),
('a03','KA031181'),
('a04','KA095477'),
('a05','KA041702');
```

	driver_id	reg_num
▶	a03	KA031181
	a05	KA041702
	a01	KA052250
	a02	KA053408
	a04	KA095477

insert into participated

```
values('a01','KA052250',11,10000),
('a02','KA053408',12,50000),
('a03','KA031181',13,25000),
('a04','KA095477',14,3000),
('a05','KA041702',15,5000);
```

	driver_id	reg_num	report_num	damage_amount
▶	a01	KA052250	11	10000
	a02	KA053408	12	50000
	a03	KA031181	13	25000
	a04	KA095477	14	3000
	a05	KA041702	15	5000

Queries

- Update the damage amount to 25000 for the car with a specific reg-num (example 'KA053408') for which the accident report number was 12.

```
update participated set damage_amount=25000
```

```
where reg_num='KA053408' and report_num=12;
```


	driver_id	reg_num	report_num	damage_amount
▶	a01	KA052250	11	10000
	a02	KA053408	12	25000
	a03	KA031181	13	25000
	a04	KA095477	14	3000
	a05	KA041702	15	5000

- Add a new accident to the database.

```
insert into accident
values (16,'2015-03-08','domlur');
```

	report_num	accident_date	location
▶	11	2001-01-03	mysoreroad
	12	2002-02-04	southendcircle
	13	2021-01-03	bulltempleroad
	14	2017-02-08	mysoreroad)
	15	2004-03-05	kanakpuraroad

- Display Accident date and location

```
select accident_date ,location from accident;
```

	accident_date	location
▶	2001-01-03	mysoreroad
	2002-02-04	southendcircle
	2021-01-03	bulltempleroad
	2017-02-08	mysoreroad)
	2004-03-05	kanakpuraroad

- Display driver id who did accident with damage amount greater than or equal to Rs.25000

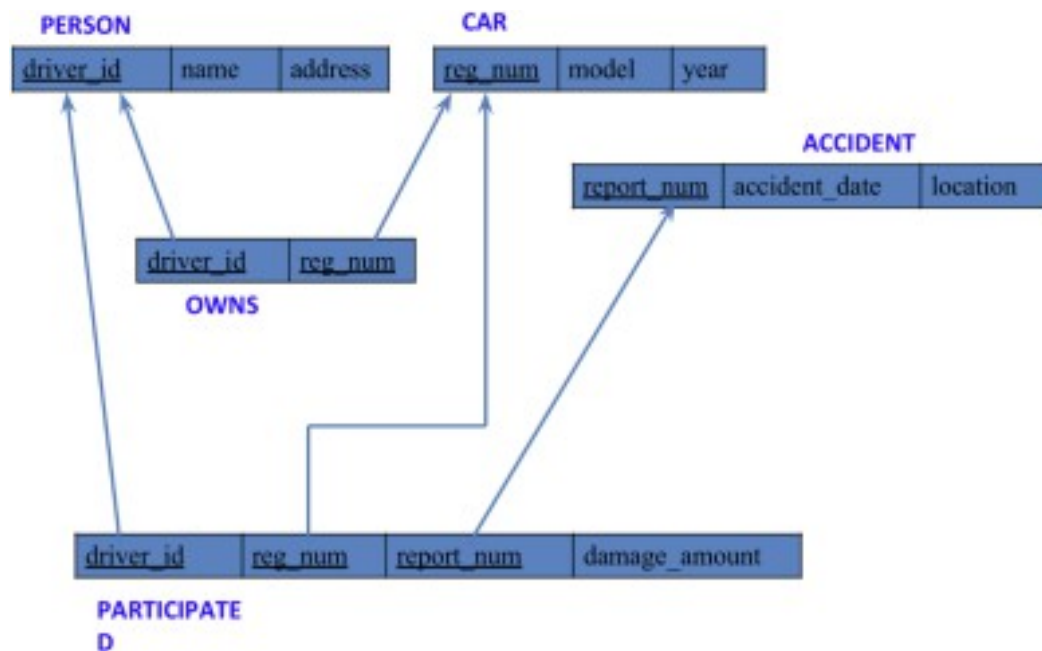
```
select driver_id from participated
where damage_amount>=25000;
```

	driver_id
▶	a02
	a03

WEEK -02

- PERSON (driver_id: String, name: String, address: String)
- CAR (reg_num: String, model: String, year: int)
- ACCIDENT (report_num: int, accident_date: date, location: String)
- OWNS (driver_id: String, reg_num: String)
- PARTICIPATED (driver_id: String, reg_num: String, report_num: int, damage_amount: int)

Schema Diagram



*select * from car
order by year asc;

Result Grid			
Filter Rows:			
	reg_num	model	year
▶	KA031181	lancer	1957
	KA052250	indica	1990
	KA095477	toyota	1998
	KA041702	audi	2005
	KA053408	honda	2008
•	NULL	NULL	NULL

```
select count(report_num)
from car c , participated p
where c.reg_num=p.reg_num and c.model='lancer';
```

Result Grid		Filter Rows:
	count(report_num)	
▶	1	

```
select count(driver_id)
from accident a,participated p
where a.report_num=p.report_num and a.accident_date like '__08%';
```

Result Grid		Filter Rows:
	count(driver_id)	
▶	0	

```
select avg(damage_amount) as 'average' from participated;
```

Result Grid		Filter Rows:
	average	
▶	13600.0000	

```
select name
from person a ,participated p
where a.driver_id=p.driver_id and p.damage_amount >(select avg(damage_amount) from participated);
```

Result Grid	
	name
▶	pradeep
	smith

```
select max(damage_amount) from participated;
```

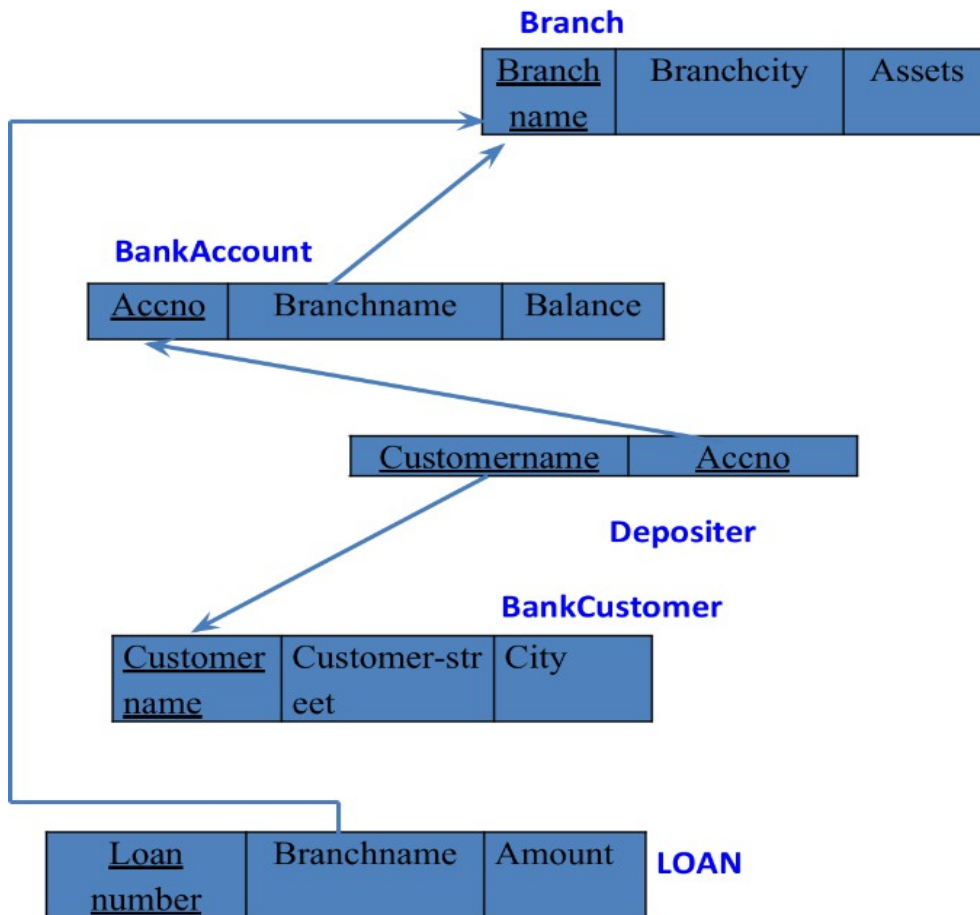
Result Grid		Filter Rows:
	max(damage_amount)	
▶	25000	

BANK ACCOUNT

QUESTION (WEEK 3)

- Create the above tables by properly specifying the primary keys and the foreign keys.
- Enter at least five tuples for each relation.
- Display the branch name and assets from all branches in lakhs of rupees and rename The assets column to 'assets in lakhs'.
- Find all the customers who have at least two accounts at the same branch (ex.SBI_residencyroad).
- Create a view which gives each branch the sum of the amount of all the loans at the branch.

SCHEMA DIAGRAM



CREATE DATABASE

```
create database Hitish_116;  
use Hitish_116;
```

CREATE TABLES

```
create table branch (  
branchname varchar(50),  
branchcity varchar(50),  
assests int ,  
primary key (branchname));
```

```
create table bankcustomer(  
customername varchar(50),  
customer_street varchar(50),  
city varchar(50),  
primary key(customername));
```

```
create table bankaccount (  
accno int,  
branchname varchar(50),  
balance int,  
primary key (accno),  
foreign key (branchname) references branch (branchname));
```

```
create table depositer(  
customername varchar(50),  
accno int,  
primary key (customername, accno),  
foreign key (customername) references bankcustomer(customername),  
foreign key (accno) references bankaccount(accno));
```

```
create table loan(  
loannumber int,  
branchname varchar(50),  
amount int,  
primary key (loannumber),  
foreign key (branchname) references branch (branchname));
```

STRUCTURE OF TABLE

```
desc branch;
```

Result Grid						
		Filter Rows:			Export:	Wrap
Field	Type	Null	Key	Default	Extra	
branchname	varchar(50)	NO	PRI	NULL		
branchcity	varchar(50)	YES		NULL		
assests	int	YES		NULL		

desc bankaccount;

Result Grid						
		Filter Rows:			Export:	Wrap Cell C
Field	Type	Null	Key	Default	Extra	
accno	int	NO	PRI	NULL		
branchname	varchar(50)	YES	MUL	NULL		
balance	int	YES		NULL		

desc depositer;

Result Grid						
		Filter Rows:			Export:	Wrap Cell
Field	Type	Null	Key	Default	Extra	
customername	varchar(50)	NO	PRI	NULL		
accno	int	NO	PRI	NULL		

desc bankcustomer;

Result Grid						
		Filter Rows:			Export:	Wrap Cell Content: fA
Field	Type	Null	Key	Default	Extra	
customername	varchar(50)	NO	PRI	NULL		
customer_street	varchar(50)	YES		NULL		
city	varchar(50)	YES		NULL		

desc loan;

Result Grid						
		Filter Rows:			Export:	Wrap Cell Content:
Field	Type	Null	Key	Default	Extra	
loannumber	int	NO	PRI	NULL		
branchname	varchar(50)	YES	MUL	NULL		
amount	int	YES		NULL		

INSERTING VALUES INTO THE TABLE

insert into branch

values('SBI-chamrajpet','banglore', 50000),

('SBI-residencyroad','banglore',10000),

('SBI-shivajiroad','bombay',20000),

```
('SBI-parliamentroad','delhi',10000),
('SBI-jantarmanatar','delhi',20000);
```

Result Grid			
Filter Rows:			
	branchname	branchcity	asests
▶	SBI-chamrajpet	banglore	50000
	SBI-jantarmanatar	delhi	20000
	SBI-parliamentroad	delhi	10000
	SBI-residencyroad	banglore	10000
	SBI-shivajiroad	bombay	20000
*	NULL	NULL	NULL

```
insert into bankcustomer
values('avinash','bull-temple-road','banglore'),
('dinesh','bannergatta-road','banglore'),
('mohan','nationalcollege-road','banglore'),
('nikil','akbar-road','delhi'),
('ravi','prithviraj-road','delhi');
```

Result Grid			
Filter Rows:			
	customername	customer_street	city
▶	avinash	bull-temple-road	banglore
	dinesh	bannergatta-road	banglore
	mohan	nationalcollege-road	banglore
	nikil	akbar-road	delhi
	ravi	prithviraj-road	delhi
*	NULL	NULL	NULL

```
insert into bankaccount
values(1,'SBI-chamrajpet',2000),
(2,'SBI-residencyroad',5000),
(3,'SBI-shivajiroad',6000),
(4,'SBI-parliamentroad',9000),
(5,'SBI-jantarmanatar',8000),
(6,'SBI-shivajiroad',4000),
(8,'SBI-residencyroad',4000),
```



```
(9,'SBI-parlimentroad',3000),
(10,'SBI-residencyroad',5000),
(11,'SBI-jantarmantar',2000);
```

Result Grid			
Filter Rows:			
	accno	branchname	balance
▶	1	SBI-chamrajpet	2000
	2	SBI-residencyroad	5000
	3	SBI-shivajiroad	6000
	4	SBI-parlimentroad	9000
	5	SBI-jantarmantar	8000
	6	SBI-shivajiroad	4000
	8	SBI-residencyroad	4000
	9	SBI-parlimentroad	3000
	10	SBI-residencyroad	5000
	11	SBI-jantarmantar	2000
•	NULL	NULL	NULL

insert into depositer

```
values('avinash',1),
('dinesh',2),
('nikil',4),
('ravi',5),
('avinash',8),
('nikil',9),
('dinesh',10),
('nikil',11);
```

Result Grid		
Filter Rows:		
	customername	accno
▶	avinash	1
	dinesh	2
	nikil	4
	ravi	5
	avinash	8
	nikil	9
	dinesh	10
	nikil	11
•	NULL	NULL

insert into loan

```
values(1,'SBI-chamrajpet',1000),
(2,'SBI-residencyroad',2000),
(3,'SBI-shivajiroad',3000),
(4,'SBI-parlimentroad',4000),
(5,'SBI-jantarmantar',5000);
```

Result Grid			
Filter Rows:			
	loannumber	branchname	amount
▶	1	SBI-chamrajpet	1000
	2	SBI-residencyroad	2000
	3	SBI-shivajiroad	3000
	4	SBI-parlimentroad	4000
	5	SBI-jantarmantar	5000
*	NULL	NULL	NULL

QUERIES

1. Display the branch name and assets from all branches in lakhs of rupees and rename the assets column to 'assets in lakhs'.

```
select branchname,assests as 'assets in lakhs'
from branch;
```

Result Grid		
Filter Rows:		
	branchname	assets in lakhs
▶	SBI-chamrajpet	50000
	SBI-jantarmantar	20000
	SBI-parlimentroad	10000
	SBI-residencyroad	10000
	SBI-shivajiroad	20000
*	NULL	NULL

2. Find all the customers who have at least two accounts at the same branch (ex.SBI_ResidencyRoad).

```
select d.customername
from bankaccount b, depositer d
where b.accno=d.accno and branchname='SBI-residencyroad'
group by customername
having count(*)>=2;
```

Result Grid	
	customername
▶	dinesh

3. Create a view which gives each branch the sum of the amount of all the loans at the branch.

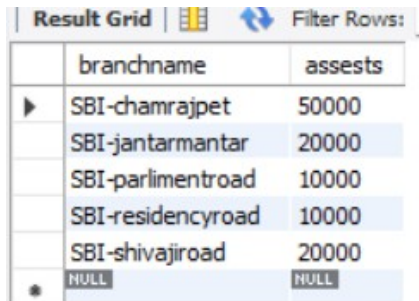
```
create view loan_info as
select b.branchname, sum(l.amount)
from branch b , loan l
where b.branchname=l.branchname
group by l.branchname;
select * from loan_info;
```

	branchname	sum(l.amount)
▶	SBI-chamrajpet	1000
	SBI-residencyroad	2000
	SBI-shivajiroad	3000
	SBI-parlimentroad	4000

WEEK -04

4. Retrieve all branches and their respective total assets

```
select branchname, assests  
from branch;
```



The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The grid contains the following data:

	branchname	assests
▶	SBI-chamrajpet	50000
	SBI-jantarmantar	20000
	SBI-parlimentroad	10000
	SBI-residencyroad	10000
	SBI-shivajiroad	20000
*	NULL	NULL

5. List all customers who live in a particular city

```
select customername  
from bankcustomer  
where city="Delhi";
```

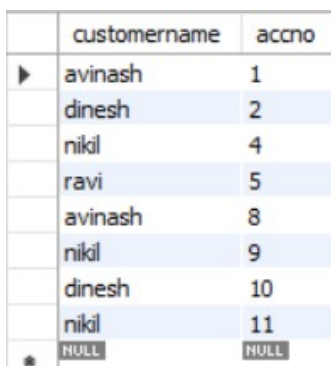


The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The grid contains the following data:

	customer_name
▶	Nikil

6. List all customers with their account numbers

```
select customername ,accno  
from depositer ;
```

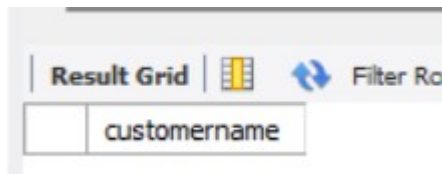


The screenshot shows a 'Result Grid' window with a 'Filter Rows' button. The grid contains the following data:

	customername	accno
▶	avinash	1
	dinesh	2
	nikil	4
	ravi	5
	avinash	8
	nikil	9
	dinesh	10
	nikil	11
*	NULL	NULL

7. Find all the customers who have an account at all the branches located in a specific city (Ex. Delhi).

```
select c.customername
from bankcustomer c, depositer d, bankaccount a, branch b
where c.customername=d.customername and d.accno=a.accno and a.branchname=b.branchname and
b.branchname=all(select b.branchname
                  from branch b
                  where b.branchcity='delhi');
```

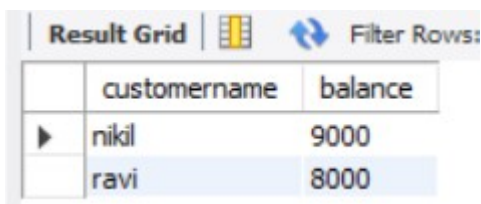


The screenshot shows a database interface with a 'Result Grid' tab. Below the tab, there is a single column header 'customername'.

customername

8. Find all customers who have accounts with a balance greater than a specified amount (5000)

```
select c.customername, b.balance
from bankcustomer c, bankaccount b, depositer d
where d.accno=b.accno and c.customername=d.customername and b.balance>5000;
```



The screenshot shows a database interface with a 'Result Grid' tab. Below the tab, there is a table with two columns: 'customername' and 'balance'. The table contains two rows of data: 'nikil' with a balance of 9000, and 'ravi' with a balance of 8000.

customername	balance
nikil	9000
ravi	8000

9. List all branch who have both a loan and an account

```
select distinct(b.branchname)
from branch b, bankaccount a, loan l
where b. branchname=a.branchname and b.branchname=l.branchname;
```

Result Grid			Filter
	branchname		
▶	SBI-chamrajpet		
	SBI-jantarmantar		
	SBI-parlimentroad		
	SBI-residencyroad		
	SBI-shivajiroad		

10. Get the number of accounts held at each branch

```
select branchname , count(*)
from bankaccount
group by branchname;
```

Result Grid			Filter Rows:
	branchname	count(*)	
▶	SBI-chamrajpet	1	
	SBI-jantarmantar	2	
	SBI-parlimentroad	2	
	SBI-residencyroad	3	
	SBI-shivajiroad	2	

11. Find all branches that have no loans issued

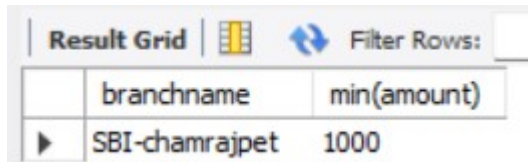
```
select b.branchname
from branch b
where b.branchname not in(select branchname
                           from loan);
```

Result Grid			
	branchname		
	NULL		

12. Retrieve the branch with the smallest total loan amount

```
select branchname ,min(amount)
```

from loan
group by branchname
order by min(amount)
limit 1;



	branchname	min(amount)
▶	SBI-chamrajpet	1000

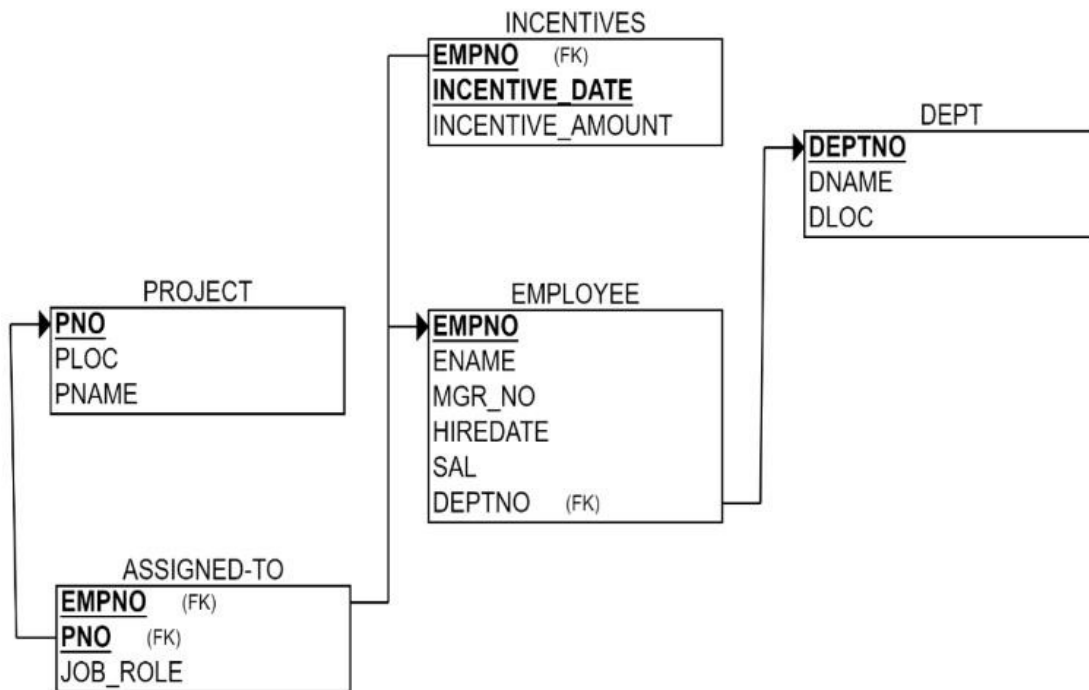
EMPLOYEE DATABASE

(WEEK -05)

QUESTION

1. Using Scheme diagram, Create tables by properly specifying the primary keys and the foreign keys.
2. Enter greater than five tuples for each table.
3. Retrieve the employee numbers of all employees who work on project located in Bengaluru, Hyderabad, or Mysuru
4. Get Employee ID's of those employees who didn't receive incentives
5. Write a SQL query to find the employees name, number, dept, job_role, department location and project location who are working for a project location same as his/her department location.

Schema Diagram



CREATE DATABASE

Create database Hitish_116;

Use database Hitish_116;

CREATE TABLES

```
create table project(  
  pno int,  
  ploc varchar(50),  
  pname varchar(50),  
  primary key (pno));
```

```
create table dept(  
  deptno int primary key,  
  dname varchar(50),  
  dloc varchar(50));
```

```
create table employee(  
  empno int primary key,  
  ename varchar(50),  
  mgr_no int,  
  hiredate date,  
  sal int,  
  deptno int foreign key references dept(deptno));
```



```
empno int primary key,
empname varchar(50),
mgr_no int,
hiredate date,
sal int,
deptno int,
foreign key (deptno) references dept (deptno));
```

```
create table incentives(
empno int ,
incentive_date date ,
incentive_amt int,
primary key(empno,incentive_date),
foreign key (empno) references employee (empno));
```

```
create table assigned_to(
empno int,
pno int,
job_role varchar (50),
primary key (empno, pno),
foreign key (empno) references employee(empno),
foreign key (pno) references project (pno));
```

STRUCTURE OF TABLE

Desc project;

	Field	Type	Null	Key	Default	Extra
►	pno	int	NO	PRI	NULL	
	ploc	varchar(50)	YES		NULL	
	pname	varchar(50)	YES		NULL	

desc dept;

Result Grid		Filter Rows:	Export:			
	Field	Type	Null	Key	Default	Extra
▶	deptno	int	NO	PRI	NULL	
	dname	varchar(50)	YES		NULL	
	dloc	varchar(50)	YES		NULL	

Desc employee;

Result Grid		Filter Rows:	Export:			
	Field	Type	Null	Key	Default	Extra
▶	empno	int	NO	PRI	NULL	
	empname	varchar(50)	YES		NULL	
	mgr_no	int	YES		NULL	
	hiredate	date	YES		NULL	
	sal	int	YES		NULL	
	deptno	int	YES	MUL	NULL	

Desc assigned_to;

Result Grid		Filter Rows:	Export:			
	Field	Type	Null	Key	Default	Extra
▶	empno	int	NO	PRI	NULL	
	pno	int	NO	PRI	NULL	
	job_role	varchar(50)	YES		NULL	

INSERT TO TABLE

insert into project

values(01,'bengaluru','pascalai'),

(02,'hyderbaad','p1m'),

(03,'mysuru','p2m'),

(04,'chennai','p3m'),

(05,'delhi','p4m');

pno	ploc	pname
1	bengaluru	pascalai
2	hyderbaad	p1m
3	mysuru	p2m
4	chennai	p3m
5	delhi	p4m

insert into dept

values(01,'sales','bengaluru'),

(02,'it','hyderabad'),

(03,'hr','mysuru'),

(04,'finance','delhi'),

(05,'production','chennai');

deptno	dname	dloc
1	sales	bengaluru
2	it	hyderabad
3	hr	mysuru
4	finance	delhi
5	production	chennai
NULL	NULL	NULL

insert into employee

values (101,"Dinesh",111,"2021-11-01",50000,1),

(102,"Dhanush",112,"2024-01-01",70000,2),

(103,"Daivya",113,"2024-01-01",80000,3),

(104,"Aditya",114,"2023-08-11",65000,1),

(105,"Arun",111,"2022-06-07",35000,2);

empno	empname	mgr_no	hiredate	sal	deptno
101	Dinesh	111	2021-11-01	50000	1
102	Dhanush	112	2024-01-01	70000	2
103	Daivya	113	2024-01-01	80000	3
104	Aditya	114	2023-08-11	65000	1
105	Arun	111	2022-06-07	35000	2
NULL	NULL	NULL	NULL	NULL	NULL

insert into incentives

values(4,'2020-11-12',3000),

(8,'2015-07-30',4000),

(7,'2010-10-14',5000),

(7,'2015-07-24',7000),

(2,'2020-11-30',3000);

empno	incentive_date	incentive_amt
101	2023-01-01	3000
101	2023-09-01	2000
102	2024-05-05	2000
104	2024-01-01	5000
105	2023-01-01	1000
NULL	NULL	NULL

```

insert into assigned_to
values(101,02,'assistant'),
(101,01,'manager'),
(102,02,'head'),
(103, 03,'manager'),
(104,05,'developer');

```

empno	pno	job_role
101	1	manager
101	2	assistant
102	2	head
103	3	manager
104	5	developer
NULL	NULL	NULL

QUERIES

1. Retrieve the employee numbers of all employees who work on project located in Bengaluru, Hyderabad, or Mysuru

```

select e.empno
from employee e, assigned_to a
where e.empno=a.empno and a.pno in(select pno
                                   from project
                                   where ploc in ('bengaluru' ,' hyderabad',' mysuru'));

```

empno	pno
101	1
103	3
NULL	NULL

2. Get Employee ID's of those employees who didn't receive incentives
select empno from employee
where empno not in (select distinct empno from incentives);

empno
103
NULL

3. Write a SQL query to find the employees name, number, dept, job_role, department location and project location who are working for a project location same as his/her department location.

```
select e.empno, e.empname, d.deptno, a.job_role, d.dloc ,p.ploc
from employee e, project p, assigned_to a, dept d
where e.empno=a.empno and p.pno=a.pno and e.deptno=d.deptno and d.dloc=p.ploc;
```

empno	empname	deptno	job_role	dloc	ploc
101	Dinesh	1	manager	bengaluru	bengaluru
103	Daivya	3	manager	mysuru	mysuru

(WEEK-06)

1. List all employees along with their project details (if assigned)

```
select e.empname, p.*  
from employee e , project p, assigned_to a  
where e.empno=a.empno and p.pno=a.pno;
```

empname	pno	ploc	pname
Dinesh	1	bengaluru	pascalai
Dinesh	2	hyderbaad	p1m
Dhanush	2	hyderbaad	p1m
Daivya	3	mysuru	p2m
Aditya	5	delhi	p4m

2. Find all employees who received incentives, along with the total incentive amount

```
select i.empno, e.empname, sum(i.incentive_amt) as total_incentive  
from incentives i, employee e  
where i.empno=e.empno  
group by empno;
```

empno	empname	total_incentive
101	Dinesh	5000
102	Dhanush	2000
104	Aditya	5000
105	Arun	1000

3. Retrieve the project names and locations of projects with employees assigned as 'Manager'

```
select p.pname, p.ploc  
from project p  
where pno in (select pno from assigned_to a where job_role='manager');
```

pname	ploc
pascalai	bengaluru
p2m	mysuru

4. List departments along with the number of employees in each department

```
select d.dname, count(e.empno)  
from dept d, employee e  
where d.deptno =e. deptno  
group by d.dname;
```

dname	count(e.empno)
sales	2
it	2
hr	1

5. Find employees who have not been assigned to any project

```
select e.empno, e.empname
from employee e
where not exists (select 1 from assigned_to a where e.empno=a.empno);
```

empno	empname
105	Arun
NULL	NULL

6. List all employees along with their department names and location

```
select e.empname , d.deptno, d.dloc
from employee e , dept d
where e.deptno=d.deptno;
```

empname	deptno	dloc
Dinesh	1	bengaluru
Dhanush	2	hyderabad
Daivya	3	mysuru
Aditya	1	bengaluru
Arun	2	hyderabad

7. Retrieve the details of employees who work under a specific manager

```
select *
from employee e
where mgr_no=111;
```

empno	empname	mgr_no	hiredate	sal	deptno
101	Dinesh	111	2021-11-01	50000	1
105	Arun	111	2022-06-07	35000	2
NULL	NULL	NULL	NULL	NULL	NULL

8. List all projects that have employees assigned and the number of employees on each project:

```
select p.pname, count(e.empno)
```

```

from project p, assigned_to e
where e.pno=p.pno
group by p.pname;

```

pname	count(e.empno)
pascalai	1
p1m	2
p2m	1
p4m	1

9. List the total number of incentives given to each employee and the sum of incentives for each

```

select empno, count(incentive_date) as number_of_times, sum(incentive_amt) as total_amt
from incentives
group by empno;

```

empno	number_of_times	total_amt
101	2	5000
102	1	2000
104	1	5000
105	1	1000

10. Retrieve all employees who have the role of 'Developer' on any project

```

select e.empno, e.empname
from employee e
where e.empno in (select empno from assigned_to where empno=e.empno and job_role='developer');

```

empno	empname
104	Aditya
NULL	NULL

11. Display the department-wise average salary of employees:

```

select deptno , avg(sal) as average
from employee
group by deptno;

```

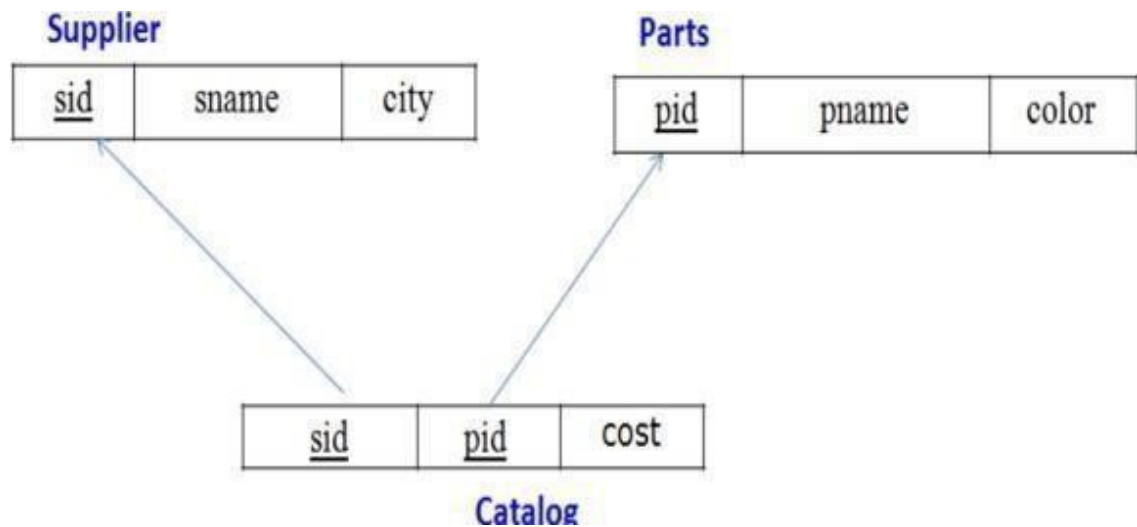
deptno	average
1	57500.0000
2	52500.0000
3	80000.0000

SUPPLIERS DATABASE

(WEEK -07) QUESTION

1. Using Scheme diagram, Create tables by properly specifying the primary keys and the foreign keys.
2. Insert appropriate records in each table.
3. Find the pnames of parts for which there is some supplier.
4. Find the snames of suppliers who supply every part.
5. Find the snames of suppliers who supply every red part.
6. Find the pnames of parts supplied by Acme Widget Suppliers and by no one else
7. Find the sids of suppliers who charge more for some part than the average cost of that part (averaged over all the suppliers who supply that part)
8. For each part, find the sname of the supplier who charges the most for that part

Schema Diagram:



Create Database:

```
create database supp;
use supp;
```

Create Tables:

```
create table Supplier(
    s_id int primary key,
    s_name varchar(30), city
```

```

varchar(30)
create table Parts( p_id int
primary key, p_name
varchar(30), color
varchar(30));

```

```

create table Catalog(
s_id int,
p_id int,
cost float,
foreign key(s_id) references Supplier(s_id),
foreign key(p_id) references Parts(p_id));

```

Structure of the Table:

desc Supplier;

	Field	Type	Null	Key	Default	Extra
►	s_id	int	NO	PRI	NULL	
	s_name	varchar(30)	YES		NULL	
	city	varchar(20)	YES		NULL	

desc Parts;

	Field	Type	Null	Key	Default	Extra
►	p_id	int	NO	PRI	NULL	
	p_name	varchar(30)	YES		NULL	
	color	varchar(30)	YES		NULL	

desc Catalog;

	Field	Type	Null	Key	Default	Extra
►	s_id	int	YES	MUL	NULL	
	p_id	int	YES	MUL	NULL	
	cost	float	YES		NULL	

Inserting Values to the tables:

```

insert into Supplier values
(10001, 'Acme_Widget', 'Bangalore'),
(10002, 'Johns', 'Kolkata'),
(10003, 'Vimal', 'Mumbai'),

```

```
(10004, 'Reliance', 'Delhi');
select * from Supplier;
```

	s_id	s_name	city
▶	10001	Acme_Widget	Bangalore
	10002	Johns	Kolkata
	10003	Vimal	Mumbai
	10004	Reliance	Delhi
•	HULL	HULL	HULL

```
insert into Parts
values (20001, 'Book',
'Red'),
(20002, 'Pen', 'Red'),
(20003, 'Pencil', 'Green'),
(20004, 'Mobile', 'Green'),
(20005, 'Charger', 'Black');
```

	p_id	p_name	color
▶	20001	Book	Red
	20002	Pen	Red
	20003	Pencil	Green
	20004	Mobile	Green
	20005	Charger	Black
•	HULL	HULL	HULL

```
insert into Catalog values
(10001, 20001, 10),
(10001, 20002, 10),
(10001, 20003, 30),
(10001, 20004, 10),
(10001, 20005, 10),
(10002, 20001, 10),
(10002, 20002, 20),
(10003, 20003, 30),
(10004, 20003, 40);
```

	s_id	p_id	cost
▶	10001	20001	10
	10001	20002	10
	10001	20003	30
	10001	20004	10
	10001	20005	10
	10002	20001	10
	10002	20002	20
	10003	20003	30
	10004	20003	40

Queries:

Find the pnames of parts for which there is some supplier.

```
select distinct p.p_name
from Supplier s, Catalog c, Parts p where
s.s_id = c.s_id and
p.p_id = c.p_id and
c.s_id is not null;
```

	p_name
▶	Book
	Pen
	Pencil
	Mobile
	Charger

Find the snames of suppliers who supply every part.

```
select distinct s_name
from Supplier s, Catalog c, Parts p
where s.s_id = c.s_id
group by s.s_id, s.s_name
having count(distinct c.p_id)=(select count(*) from Parts p);
```

	s_name
▶	Acme_Widget

Find the snames of suppliers who supply every red part.

```
select distinct s_name
from Supplier s, Catalog c, Parts p
where s.s_id = c.s_id and
c.p_id in (select p_id from Parts p where p.color = 'Red')
```

	s_name
▶	Johns
	Acme_Widget

Find the pnames of parts supplied by Acme Widget Suppliers and by no one else

```
select distinct p_name from Supplier s, Parts p, Catalog c where
p.p_id in (select c.p_id from Catalog c, Supplier s where
s.s_id = c.s_id and s.s_name = 'Acme_Widget') and
p.p_id not in (select c.p_id from Catalog c, Supplier s where
s.s_id = c.s_id and s.s_name != 'Acme_Widget');
```

	p_name
▶	Mobile
	Charger

Find the sids of suppliers who charge more for some part than the average cost of that part (averaged over all the suppliers who supply that part)

```
create view Average(p_id, Average_Product_Cost) as
select c.p_id, avg(cost)
from Catalog c
group by
c.p_id;
select c.s_id from Catalog c, Average a where
c.p_id = a.p_id and
c.cost > (a.Average_Product_Cost)
group by c.p_id, c.s_id;
```

	s_id
▶	10002
	10004

For each part, find the sname of the supplier who charges the most for that part

```
select distinct s.s_name, c.cost, c.p_id from Catalog c, Supplier s where  
s.s_id = c.s_id and  
c.cost in (select max(cost) from Catalog c group by c.p_id);
```

	sname
►	Acme Widget
	Johns
	Reliance

NO SQL STUDENT DATABASE

(WEEK -08) QUESTION

- Perform the following DB operations using MongoDB.
- Create a database “Student” with the following attributes Rollno, Age, ContactNo, Email-Id.
- Insert appropriate values
- Write query to update Email-Id of a student with rollno 10.
- Replace the student name from “ABC” to “FEM” of rollno 11.

Create Database:

db.createCollection("Student");

```
Atlas atlas-cci5oy-shard-0 [primary] test> db.createCollection("Student");
{ ok: 1 }
Atlas atlas-cci5oy-shard-0 [primary] test>
```

Inserting Values to the tables:

db.Student.insert({RollNo:1, Age:21, Cont:9876, email:"antara.de9@gmail.com"});

```
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("675fe28cf2355f925cc449c9") }
}
```

db.Student.insert({RollNo:2, Age:22, Cont:9976, email:"anushka.de9@gmail.com"});

```
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("675fe295f2355f925cc449ca") }
}
```

db.Student.insert({RollNo:3, Age:21, Cont:5576, email:"anubhav.de9@gmail.com"});

```
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("675fe29df2355f925cc449cb") }
}
```

db.Student.insert({RollNo:4, Age:20, Cont:4476, email:"pani.de9@gmail.com"});

```
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("675fe2a5f2355f925cc449cc") }
}
```

```
db.Student.insert({RollNo:10,Age:23,Cont:2276,email:"rekha.de9@gmail.com"});
```

```
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("675fe2abf2355f925cc449cd") }
}
```

Queries:

```
db.Student.find()
```

```
Atlas atlas-cci5oy-shard-0 [primary] test> db.Student.find()
[
  {
    _id: ObjectId("6746b3bd3524069968624499"),
    RollNo: 1,
    Age: 21,
    Cont: 9876,
    email: 'antara.de9@gmail.com'
  },
  {
    _id: ObjectId("6746b3c7352406996862449a"),
    RollNo: 2,
    Age: 22,
    Cont: 9976,
    email: 'anushka.de9@gmail.com'
  },
  {
    _id: ObjectId("6746b3d0352406996862449b"),
    RollNo: 3,
    Age: 21,
    Cont: 5576,
    email: 'anubhav.de9@gmail.com'
  },
  {
    _id: ObjectId("6746b3d8352406996862449c"),
    RollNo: 4,
    Age: 20,
    Cont: 4476,
    email: 'pani.de9@gmail.com'
  },
  {
    _id: ObjectId("6746b3e1352406996862449d"),
    RollNo: 10,
    Age: 23,
    Cont: 2276,
    email: 'Abhinav@gmail.com'
  },
]
```

Write query to update Email-Id of a student with rollno 10.

```
db.Student.update({RollNo:10},{ $set: {email:"Abhinav@gmail.com"}})
```

```
DeprecationWarning: Collection.update() is deprecated. Use updateOne, updateMany, or bulkWrite.
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 0,
  upsertedCount: 0
}
```


Replace the student name from “ABC” to “FEM” of rollno 11.

```
db.Student.insert({RollNo:11,Age:22,Name:"ABC",Cont:2276,email:"rea.de9@gmail. c om"});
```

```
{
  acknowledged: true,
  insertedIds: { '0': ObjectId("675fe2cbf2355f925cc449ce") }
}
```

```
db.Student.update({RollNo:11,Name:"ABC"},{$set:{Name:"FEM"}})
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
{
  _id: ObjectId("6746b419352406996862449e"),
  RollNo: 11,
  Age: 22,
  Name: 'FEM',
  Cont: 2276,
  email: 'rea.de9@gmail.com'
},
```

Import a given csv dataset from local file system into mongodb collectio

_id	RollNo	Age	Cont	email	Name
6746b6c4f73fea43f1	1	21	9876	antara.de9@gmail.com	
6746b6cbf73fea43f1	2	22	9976	anushka.de9@gmail.com	
6746b6d2f73fea43f1	3	21	5576	anubhav.de9@gmail.com	
6746b6d8f73fea43f1	4	20	4476	pani.de9@gmail.com	
6746b6def73fea43f1	10	23	2276	Abhinav@gmail.com	
6746b710f73fea43f1	11	22	2276	rea.de9@gmail.com	FEM

NO SQL CUSTOMERS DATABASE

(WEEK -09)

QUESTION

- Create a collection by name Customers with the following attributes. Cust_id, Acc_Bal, Acc_Type
- Insert at least 5 values into the table
- Write a query to display those records whose total account balance is greater than 1200 of account type 'Z' for each customer_id.
- Determine Minimum and Maximum account balance for each customer_id.
- Export the created collection into local file system
- Drop the table
- Import a given csv dataset from local file system into mongodb collection.

Create Database:

```
db.createCollection("Customer");
```

```
{ ok: 1 }
```

Inserting Values to the tables:

```
db.Customer.insertMany([ {custid: 1, acc_bal:10000, acc_type:"Saving"},  
  {custid: 1, acc_bal:20000, acc_type: "Checking"}, {custid: 3,  
  acc_bal:50000, acc_type: "Checking"}, {custid: 4, acc_bal:10000,  
  acc_type: "Saving"}, {custid: 5, acc_bal:2000, acc_type: "Checking"}]);
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId("675fe7b5f2355f925cc449cf"),  
    '1': ObjectId("675fe7b5f2355f925cc449d0"),  
    '2': ObjectId("675fe7b5f2355f925cc449d1"),  
    '3': ObjectId("675fe7b5f2355f925cc449d2"),  
    '4': ObjectId("675fe7b5f2355f925cc449d3")  
  }  
}
```

Queries:

Write a query to display those records whose total account balance is greater than 1200 of account type 'Z' for each customer_id.

```
db.Customer.find({acc_bal: {$gt: 12000}, acc_type:"Checking"});
```

```
[
  {
    _id: ObjectId("675fe7b5f2355f925cc449d0"),
    custid: 1,
    acc_bal: 20000,
    acc_type: 'Checking'
  },
  {
    _id: ObjectId("675fe7b5f2355f925cc449d1"),
    custid: 3,
    acc_bal: 50000,
    acc_type: 'Checking'
  }
]
```

Determine Minimum and Maximum account balance for each customer_id.

```
db.Customer.aggregate([{$group: {_id:"$custi
```

```
[
  { _id: 3, minBal: 50000, maxBal: 50000 },
  { _id: 5, minBal: 2000, maxBal: 2000 },
  { _id: 1, minBal: 10000, maxBal: 20000 },
  { _id: 4, minBal: 10000, maxBal: 10000 }
]
```

```
d", minBal:{$min:"$acc_bal"}, maxBal:{$max:"$acc_bal"}}]);
```

```
db.Customers.drop()
```

```
true
```

Import a given csv dataset from local file system into mongodb collection.

_id	custid	acc_bal	acc_type
674ff20946b4cd1ffe	1	10000	Saving
674ff20946b4cd1ffe	1	20000	Checking
674ff20946b4cd1ffe	3	50000	Checking
674ff20946b4cd1ffe	4	10000	Saving
674ff20946b4cd1ffe	5	2000	Checking

NO SQL RESTAURANTS DATABASE

(WEEK-10) QUESTION

- Write a MongoDB query to display all the documents in the collection restaurants.
- Write a MongoDB query to arrange the name of the restaurants in descending along with all the columns.
- Write a MongoDB query to find the restaurant Id, name, town and cuisine for those restaurants which achieved a score which is not more than 10.
- Write a MongoDB query to find the average score for each restaurant.
- Write a MongoDB query to find the name and address of the restaurants that have a zipcode that starts with '10'.

Create Database:

```
db.createCollection("restaurants");
```

```
{ ok: 1 }
```

Inserting Values to the tables:

```
db.restaurants.insertMany([ { name: "Meghna Foods", town: "Jayanagar", cuisine: "Indian", score: 8, address: { zipcode: "10001", street: "Jayanagar" } }, { name: "Empire", town: "MG Road", cuisine: "Indian", score: 7, address: { zipcode: "10100", street: "MG Road" } }, { name: "Chinese WOK", town: "Indiranagar", cuisine: "Chinese", score: 12, address: { zipcode: "20000", street: "Indiranagar" } }, { name: "Kyotos", town: "Majestic", cuisine: "Japanese", score: 9, address: { zipcode: "10300", street: "Majestic" } }, { name: "WOW Momos", town: "Malleshwaram", cuisine: "Indian", score: 5, address: { zipcode: "10400", street: "Malleshwaram" } } ])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId("67600441f2355f925cc449d4"),
    '1': ObjectId("67600441f2355f925cc449d5"),
    '2': ObjectId("67600441f2355f925cc449d6"),
    '3': ObjectId("67600441f2355f925cc449d7"),
    '4': ObjectId("67600441f2355f925cc449d8")
  }
}
```

Queries:

Write a MongoDB query to display all the documents in the collection restaurants.

```
db.restaurants.find({})
```

```
[
  {
    _id: ObjectId("67600441f2355f925cc449d4"),
    name: 'Meghna Foods',
    town: 'Jayanagar',
    cuisine: 'Indian',
    score: 8,
    address: { zipcode: '10001', street: 'Jayanagar' }
  },
  {
    _id: ObjectId("67600441f2355f925cc449d5"),
    name: 'Empire',
    town: 'MG Road',
    cuisine: 'Indian',
    score: 7,
    address: { zipcode: '10100', street: 'MG Road' }
  },
  {
    _id: ObjectId("67600441f2355f925cc449d6"),
    name: 'Chinese WOK',
    town: 'Indiranagar',
    cuisine: 'Chinese',
    score: 12,
    address: { zipcode: '20000', street: 'Indiranagar' }
  },
  {
    _id: ObjectId("67600441f2355f925cc449d7"),
    name: 'Kyotos',
    town: 'Majestic',
    cuisine: 'Japanese',
    score: 9,
    address: { zipcode: '10300', street: 'Majestic' }
  },
  {
    _id: ObjectId("67600441f2355f925cc449d8"),
    name: 'WOW Momos',
    town: 'Malleshwaram',
    cuisine: 'Indian',
    score: 5,
    address: { zipcode: '10400', street: 'Malleshwaram' }
  }
]
```

Write a MongoDB query to arrange the name of the restaurants in descending along with all the columns

```
db.restaurants.find({}).sort({ name: -1 })
```

```
[
  {
    _id: ObjectId("67600441f2355f925cc449d8"),
    name: 'WOW Momos',
    town: 'Malleshwaram',
    cuisine: 'Indian',
    score: 5,
    address: { zipcode: '10400', street: 'Malleshwaram' }
  },
  {
    _id: ObjectId("67600441f2355f925cc449d4"),
    name: 'Meghna Foods',
    town: 'Jayanagar',
    cuisine: 'Indian',
    score: 8,
    address: { zipcode: '10001', street: 'Jayanagar' }
  },
  {
    _id: ObjectId("67600441f2355f925cc449d7"),
    name: 'Kyotos',
    town: 'Majestic',
    cuisine: 'Japanese',
    score: 9,
    address: { zipcode: '10300', street: 'Majestic' }
  },
  {
    _id: ObjectId("67600441f2355f925cc449d5"),
    name: 'Empire',
    town: 'MG Road',
    cuisine: 'Indian',
    score: 7,
    address: { zipcode: '10100', street: 'MG Road' }
  },
  {
    _id: ObjectId("67600441f2355f925cc449d6"),
    name: 'Chinese WOK',
    town: 'Indiranagar',
    cuisine: 'Chinese',
    score: 12,
    address: { zipcode: '20000', street: 'Indiranagar' }
  }
]
```

Write a MongoDB query to find the restaurant Id, name, town and cuisine for those restaurants which achieved a score which is not more than 10.

```
db.restaurants.find({ "score": { $lte: 10 } }, { _id: 1, name: 1, town: 1, cuisine: 1 })
```

```
[
  {
    _id: ObjectId("67600441f2355f925cc449d4"),
    name: 'Meghna Foods',
    town: 'Jayanagar',
    cuisine: 'Indian'
  },
  {
    _id: ObjectId("67600441f2355f925cc449d5"),
    name: 'Empire',
    town: 'MG Road',
    cuisine: 'Indian'
  },
  {
    _id: ObjectId("67600441f2355f925cc449d7"),
    name: 'Kyotos',
    town: 'Majestic',
    cuisine: 'Japanese'
  },
  {
    _id: ObjectId("67600441f2355f925cc449d8"),
    name: 'WOW Momos',
    town: 'Malleshwaram',
    cuisine: 'Indian'
  }
]
```

Write a MongoDB query to find the average score for each restaurant.

```
db.restaurants.aggregate([ { $group: { _id: "$name", average_score: { $avg: "$score" } } }])
```

```
[
  { _id: 'Meghna Foods', average_score: 8 },
  { _id: 'Kyotos', average_score: 9 },
  { _id: 'Chinese WOK', average_score: 12 },
  { _id: 'WOW Momos', average_score: 5 },
  { _id: 'Empire', average_score: 7 }
]
```

Write a MongoDB query to find the name and address of the restaurants that have a zipcode that starts with '10'.

```
db.restaurants.find({ "address.zipcode": /^10/ }, { name: 1, "address.street": 1, _id: 0 })
```

```
[
  { name: 'Meghna Foods', address: { street: 'Jayanagar' } },
  { name: 'Empire', address: { street: 'MG Road' } },
  { name: 'Kyotos', address: { street: 'Majestic' } },
  { name: 'WOW Momos', address: { street: 'Malleshwaram' } }
]
```